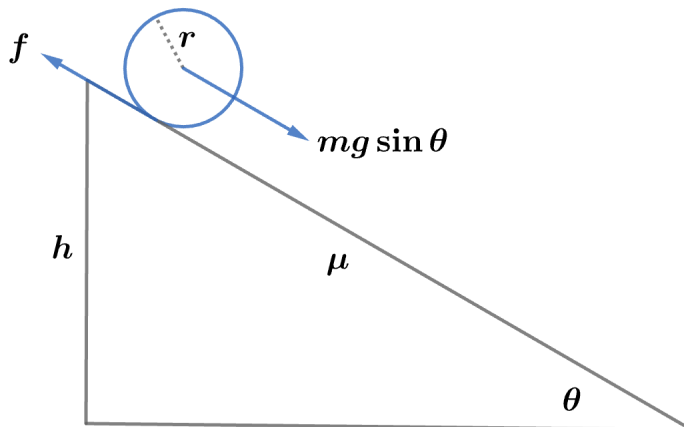


2015 F=ma Exam: Problem 22

Kevin S. Huang



Recall the acceleration of an object rolling without slipping down an incline is given by

$$a = \frac{g \sin \theta}{1 + \beta}$$

where the object has moment of inertia $I = \beta mr^2$. For small enough θ , static friction is large enough to provide sufficient torque since

$$\tau = fr = I\alpha = \beta mr^2 \alpha$$

Using $a = r\alpha$ as the object is rolling without slipping,

$$f = \beta ma = \frac{\beta mg \sin \theta}{1 + \beta}$$

so we see that the required friction force $f \rightarrow 0$ as $\theta \rightarrow 0$. Thus, the object rolls without slipping (no energy loss) for some range of angles $0 < \theta < \theta_c$.

On the other end, if $\theta = 90^\circ$ then the object drops vertically and there is also no dissipation. In between when $\theta_c < \theta < 90^\circ$, the object is rolling with slipping so there is energy dissipation. Going a bit past θ_c leads to a small amount of slipping (and energy loss) since the friction force changes continuously. Hence, the only graph that accounts for all these features is C.